

Introduction to Robotics

Robotics is a branch of engineering and science that deals with the design, construction, operation, and application of robots. It combines Mechanical Engineering, Electrical Engineering, and Computer Science. **What is a Robot?**

A robot is a programmable machine that can sense its environment, process information, and perform actions automatically or with minimal human intervention. **Components**

Sensors, Controller, Actuators, Power Supply, End Effector. **Types**

Industrial, Service, Mobile, Humanoid, Medical. **Applications**

Manufacturing, Healthcare, Agriculture, Space Exploration, Defense, Education. **Advantages**

High accuracy, productivity, safety, continuous operation. **Challenges**

High cost, programming complexity, safety and ethical concerns. **Conclusion**

Robotics is transforming modern industries through automation and intelligent machines.